EER Amplitude Modulator for Raspberry Pi GPIO4

Analysis of the SoapySDR_rpitx project and a linear, transistor-only, envelope-controlled attenuator for 200 MHz (500 MHz variant)

Date: 5 November 2026 · Scope: analysis, schematic, bill of materials, bring-up plan · Status: preliminary design

1. Executive summary

This report analyses the SoapySDR_rpitx driver and proposes an analog amplitude path for Raspberry Pi GPIO4 that replaces the pad-drive amplitude method with a continuous, linear control. The RF carrier stays a full-swing square wave at fixed pad drive, generated by librpitx. The EER envelope arrives as a PWM signal on a second GPIO pin, is low-pass filtered, and sets the tail current of a switched-current pair. The pair steers that current into a tuned tank, so the fundamental output amplitude is proportional to the envelope.

The circuit uses three transistors and only resistors, capacitors and inductors. There are no integrated circuits, and the 200 MHz build uses through-hole parts only. The design is laid out for 200 MHz. Section 8 lists the component changes needed for 500 MHz. Estimated full-scale output is about +8.5 dBm into 50 Ω, and the estimated third harmonic is about -44 dBc at 200 MHz. These are first-order calculations and must be confirmed on the bench. The most important open issue is carrier feedthrough through the collector-base capacitance of the RF switch (Section 10): it is expected to limit the usable amplitude range to roughly 10 to 15 dB until it is reduced, which is well short of what a full EER system needs.

Parameter	200 MHz build	500 MHz variant (Section 8)
Carrier source	librpitx on GPIO4 (-c 1), fixed maximum pad drive	same
Amplitude control	Analog; 9-bit PWM resolution	same
Envelope bandwidth (-3 dB)	about 1.7 kHz (filter as drawn)	scale RC values for a wider envelope
Output, full scale, 50 Ω	about +8.5 dBm (collector swing limited by the 5 V supply)	about +8.5 dBm if the switch device is fast enough
Usable amplitude range	about 10 to 15 dB until feedthrough is reduced (Section 10)	same, to be measured
Estimated 3rd harmonic	about -44 dBc	about -47 dBc
Supply	+5 V, about 14 mA peak; +3.3 V, about 4 mA	same
Active devices	1 × 2N3904, 2 × 2N5179 (matched pair)	1 × 2N3904, 2 × BFR91A (matched pair)
Other parts	resistors, capacitors, two inductors, one trimmer	same, new values

2. Analysis of the SoapySDR_rpitx project

SoapySDR_rpitx is a SoapySDR driver module that wraps librpitx, the Raspberry Pi transmit library by F5OEO. It lets SoapySDR-based software, including GNU Radio flowgraphs, use a Raspberry Pi GPIO pin as a transmitter. The repository is licensed under GPL-3.0. At the time of review it had 18 commits on the main branch, two stars, one fork and one open issue.

The README describes the project as "WIP!!!! Working but unstable. A lot of glitches". Build instructions cover compiling in the build directory and copying the resulting libSoapySDR_rpitx.so into the SoapySDR modules path, with a warning that the path depends on the SoapySDR version. The repository has no API reference, no list of supported stream formats or parameters, and no description of how amplitude or GPIO selection is exposed. A temp/ folder is committed to the tree, which suggests scratch material was included.

Upstream context helps interpret the code. The mirrored rpitx documentation (reference 3) lists a carrier range of 130 kHz to 750 MHz, GPIO18 (pin 12) as the default output, a -c 1 option that moves the output to GPIO4 (pin 7), and an RFA mode that takes an amplitude value; newer upstream releases may state a wider range, so check the version you run. The gr-rpitx paper states that librpitx added amplitude tuning on top of the original PiFm-style method, which is the origin of the amplitude capability the driver relies on.

What I could and could not verify. I could read the repository front page and the upstream documentation. GitHub blocks automated access to source-tree pages, so I could not open src/ or librpitx/ directly. The statements about the amplitude mechanism in this report therefore rest on your description, which I checked for plausibility rather than against the code. The claim of eight drive levels plus off fits the pad hardware of the Broadcom BCM283x family as I understand it, where a 3-bit drive field selects 2 to 16 mA in 2 mA steps. Please confirm the exact mapping in the datasheet for your SoC revision and in the librpitx source.

Assessment. The driver is a reasonable transmit path for carriers and for constant-envelope modulation. Its weakest point for linear work is the amplitude mechanism itself, because amplitude is set by pad drive rather than by a continuous control. An external analog stage supplies that control without changing the driver.

Peripheral conflict to check. The envelope PWM uses the PWM0 output on GPIO12. librpitx may pace its DMA transfers from the PWM or from the PCM peripheral, depending on the mode; this could not be checked in the source. If the PWM peripheral is already used for pacing, the envelope PWM cannot share it, and either the pacing must be moved to PCM or the envelope must be produced another way. The PWM clock must also be set to 19.2 MHz (with a range of 512) to obtain the 37.5 kHz used in this design. On Raspberry Pi 1 to 3 this is the crystal frequency; the Pi 4 crystal runs at 54 MHz, so there the 19.2 MHz clock has to come from a divided PLL source, and the clock settings must be checked against the code in use.

3. Why pad-drive amplitude is a poor EER amplitude path

Quantisation. EER amplitude restoration needs a continuous level. Eight drive steps plus off give about three bits of resolution. That is coarse for an envelope and produces quantisation products that land in the carrier neighbourhood.

Non-uniform steps. Drive strength sets the pad output resistance, and the upper levels saturate, so the amplitude-to-code map is nonlinear. A lookup table can make the map monotonic, but it cannot add resolution that the hardware does not have.

Edge rate and AM-to-PM conversion. Changing drive also changes the rise and fall times of the carrier edges. An amplitude change therefore shifts edge timing, which lands directly in the phase path. This is AM-to-PM conversion, and it corrupts the phase signal that EER depends on.

Shared-bank noise. Switching pad drive changes simultaneous switching current on the GPIO bank, which couples into neighbouring pins and into the supply.

Conclusion. The proposed approach keeps GPIO4 at fixed maximum drive so the carrier edges are constant. It moves amplitude control into an analog stage driven by a PWM pin. This is the attenuator approach described in the brief.

4. Architecture and how the circuit works

The phase path (librpitx writing the carrier to GPIO4) and the amplitude path (this circuit) are kept separate. The amplitude path is a 9-bit PWM signal, filtered to a slowly varying voltage and scaled by a resistor network into the base voltage V_C of Q1. V_C sets a tail current I_{tail} through Q1. Q2 and Q3 form a switched-current pair. The RF square wave on GPIO4 directs I_{tail} into the tank when Q2 conducts, and into a dummy load (R12) when Q3 conducts. The fundamental of a 50 % duty current square wave is $(2/\pi)\cdot I_{tail}$, so the tank voltage at the carrier frequency is proportional to the tail current and therefore to the envelope.

In attenuator terms, the gain from the RF switch to the tank is set by the envelope. The square wave is always full swing, so the switch never sits in a partly driven state. The only analog variable is I_{tail}, and it is the single control point for amplitude.

Why the stage is linear. Above the base-emitter threshold, Q1 converts voltage to current with a transconductance of about 1/R8. The scaling network (R3, R4, R5, R6) is chosen so that V_C is about 0.65 V with the PWM at 0 %, which is the base-emitter voltage of Q1 at a few hundred microamps. That removes most of the dead zone that a bare emitter follower would have. Because the base-emitter law is logarithmic, the current is not exactly linear in V_C near the bottom of the range, and a residual tail current of about 0.3 mA remains at 0 %. The remaining errors are this curvature, a static gain error from base current and V_{BE} mismatch, and temperature drift. A calibration table corrects them (Section 9). Figure 1 shows the system; the schematic follows on Sheets 1 and 2.

Tank and output filter. The tank and the output low-pass are designed together. At 200 MHz the filter does not present 50 Ω to the tap: the real part of its input impedance is about 120 Ω, because a Butterworth filter is matched only near DC. The tap is therefore placed at 80 % of the winding, counted from the supply end, so that the load seen at the collector is about 185 Ω. This keeps the collector swing at full scale near 1.6 V peak, which the

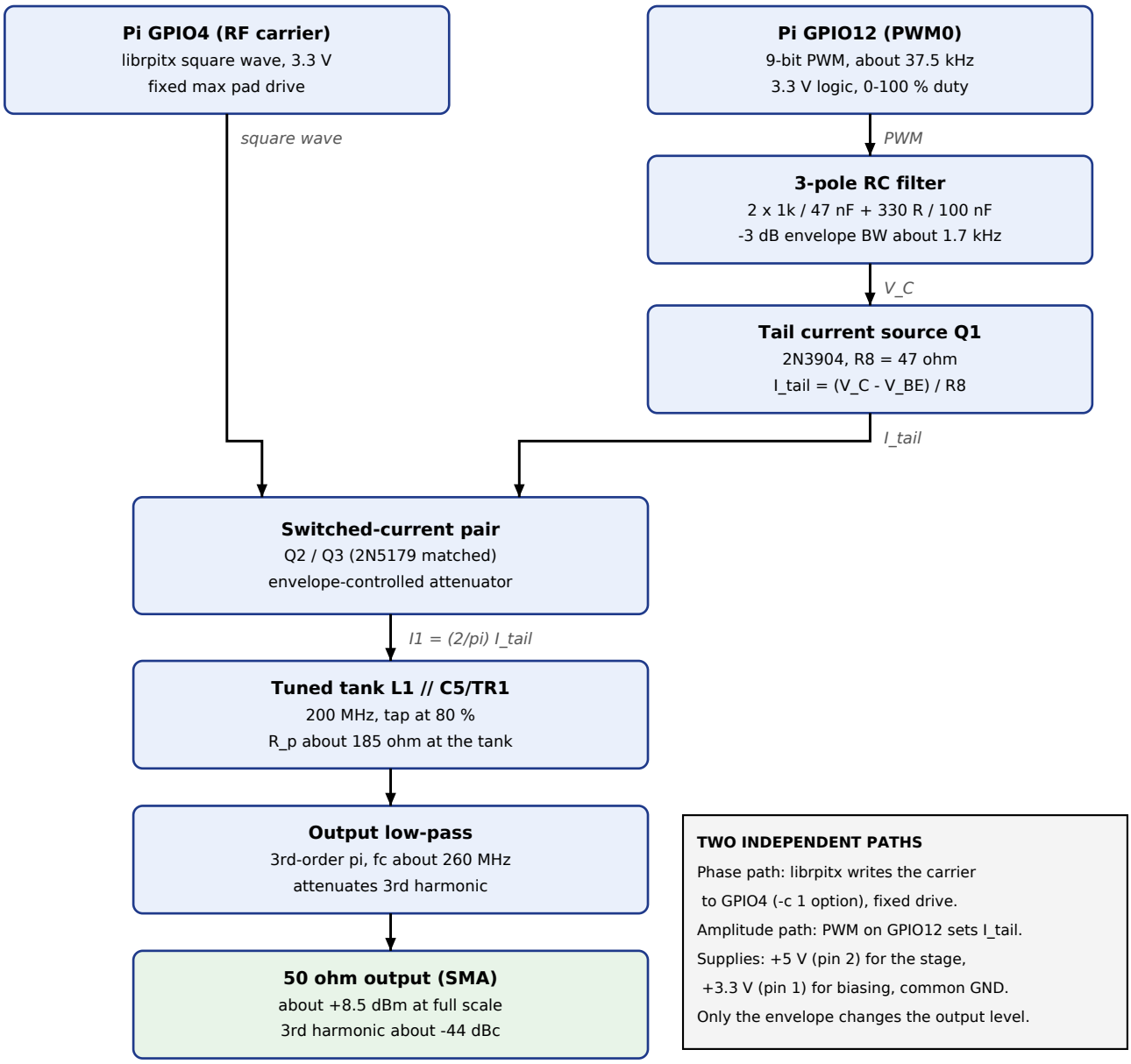
5 V supply can support (Section 6).

5. Schematic

The schematic is drawn on two A4 landscape sheets. Sheet 1 contains the envelope path: the PWM filter, the scaling network and the tail transistor. Sheet 2 contains the RF switched-current pair, the tank, the output low-pass and the supply decoupling. Nets that cross between the sheets are labelled E, +3V3 and +5V. J1 is a single connector on the Pi GPIO header (pins 1, 2, 6, 7 and 32) and is drawn in two parts, one on each sheet. Figure 1 (below) shows the block diagram; the schematic sheets follow it. Reference designators match the bill of materials in Section 7.

FIGURE 1 - System block diagram (200 MHz build)

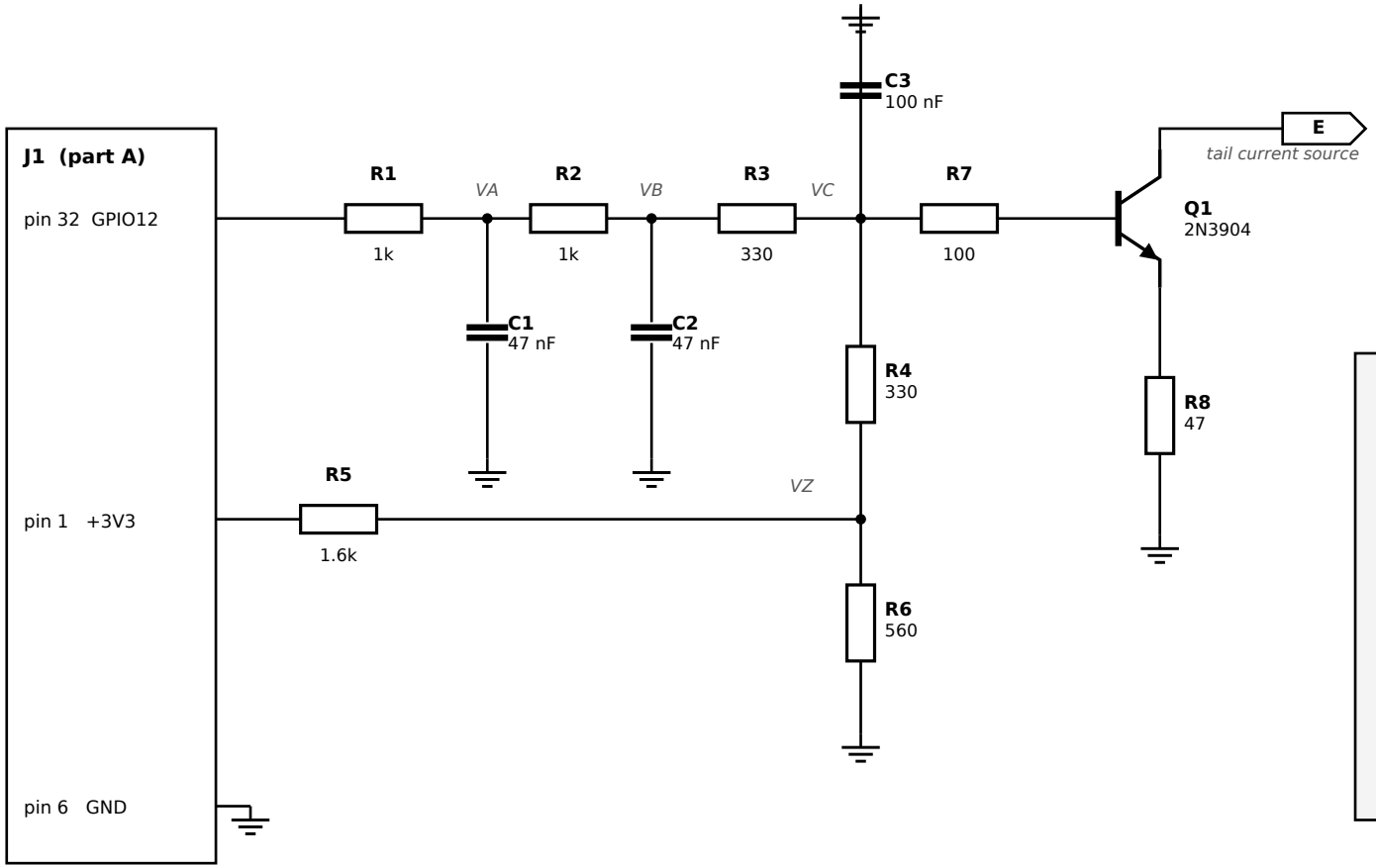
Phase path and amplitude path are separate; the RF carrier keeps a fixed pad drive



Amplitude is controlled by one analog variable (I_tail). The switching waveform stays full swing.

SHEET 1 / 2 ENVELOPE PATH

PWM from Raspberry Pi -> 3-pole RC filter -> scaling network -> tail current source Q1 (V-to-I)



NOTES - SHEET 1

Envelope scaling (loaded network): V_C = 0.242 x V_PWM + 0.649 V
- I_b x 0.66 k; V_Zth = 0.856 V, R5 || R6 = 415 ohm

Tail current: V_C - I_b(R7 + R_th) = V_BE1 + I_tail x R8
V_BE1 = 0.63 V at 0.3 mA, 0.73 V at 13.9 mA
about 0.3 mA at 0 %; 6.5 mA at 50 %; 13.9 mA at 100 %

Filter: 1k/47 nF, 1k/47 nF, 330 R/100 nF
-3 dB envelope bandwidth about 1.7 kHz (loaded)

PWM about 37.5 kHz (19.2 MHz / 512): about 48 dB ripple cut

Q1 stays active (V_CE >= 0.5 V) for the whole range

J1 is the single Pi header: pins 1, 2, 6, 7, 32 (see Sheet 2)

EER amplitude path - 200 MHz build

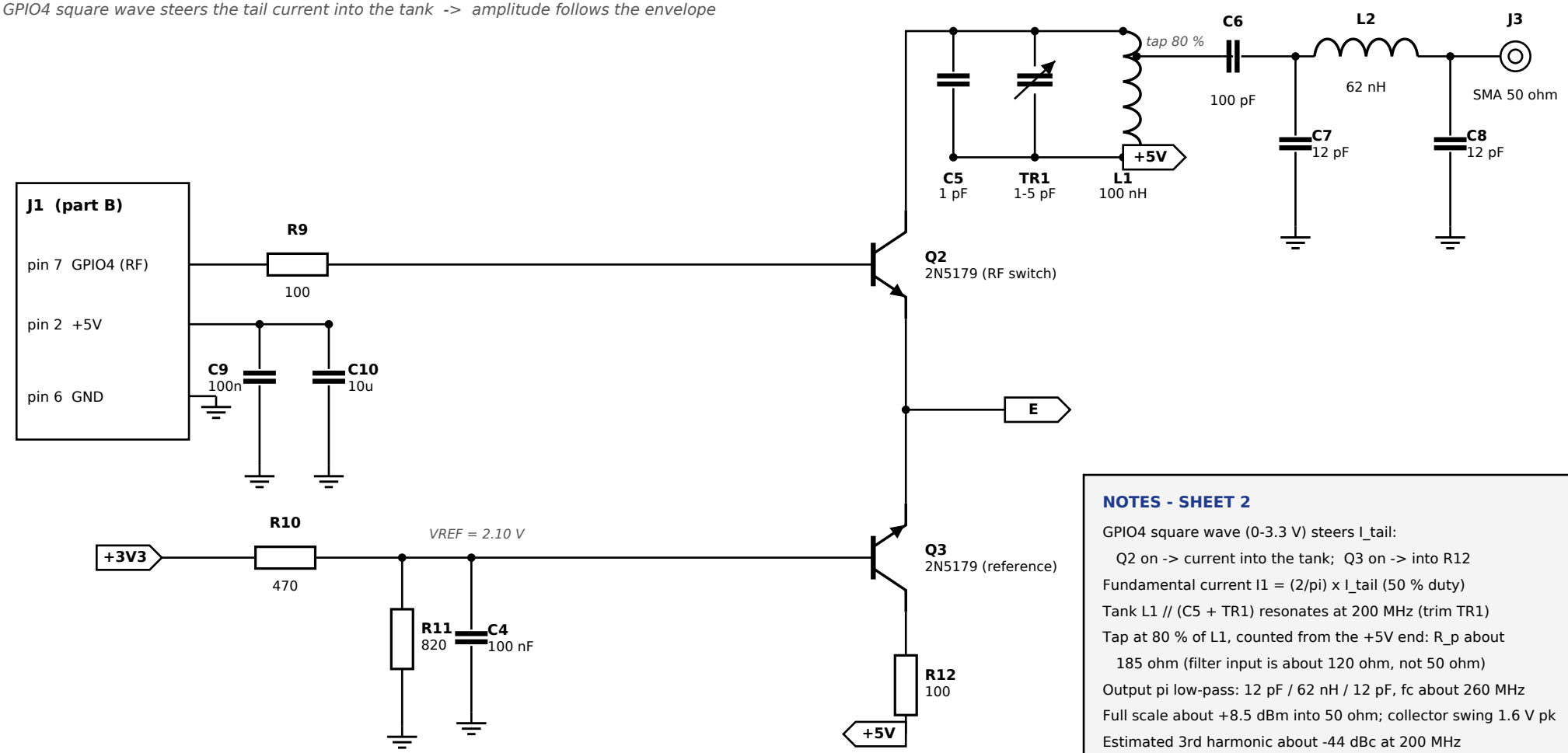
Sheet 1 of 2

Rev B - 5 Nov 2026 - preliminary, verify on bench

GND = signal ground

SHEET 2 / 2 RF SWITCHED-CURRENT PAIR, TANK AND OUTPUT FILTER

GPIO4 square wave steers the tail current into the tank -> amplitude follows the envelope



NOTES - SHEET 2

GPIO4 square wave (0-3.3 V) steers I_tail:
  Q2 on -> current into the tank; Q3 on -> into R12
Fundamental current I1 = (2/pi) x I_tail (50 % duty)
Tank L1 // (C5 + TR1) resonates at 200 MHz (trim TR1)
Tap at 80 % of L1, counted from the +5V end: R_p about
  185 ohm (filter input is about 120 ohm, not 50 ohm)
Output pi low-pass: 12 pF / 62 nH / 12 pF, fc about 260 MHz
Full scale about +8.5 dBm into 50 ohm; collector swing 1.6 V pk
Estimated 3rd harmonic about -44 dBc at 200 MHz
Q2/Q3 must be a matched pair; keep RF leads < 5 mm

EER amplitude path - 200 MHz build

Sheet 2 of 2

Rev B - 5 Nov 2026 - preliminary, verify on bench

GND = signal ground

6. Operating points and design equations

The table gives DC and first-order RF values at three PWM duty settings. The base-emitter drops follow the exponential device law (Q1: about 0.63 V at 0.3 mA, 0.71 V at 6.5 mA and 0.73 V at 13.9 mA; 2N5179: about 0.83 V at 6.5 mA and 0.88 V at 13.9 mA, from SPICE model parameters), and base currents assume a current gain of 150 for Q1 and 60 for Q2 and Q3.

Quantity	PWM 0 %	PWM 50 %	PWM 100 %
V_B (filtered PWM, after R2)	0.56 V	1.11 V	1.67 V
V_Z (R5/R6 node)	0.74 V	0.95 V	1.16 V
V_C (Q1 base node)	0.65 V	1.02 V	1.40 V
I_{tail} (through R8)	0.29 mA	6.5 mA	13.9 mA
Q1 V_{CE} at worst case	1.33 V	0.93 V	0.52 V
Fundamental current I_1	0.18 mA pk	4.14 mA pk	8.84 mA pk
Tank voltage V_1 (R_p = 184 Ω)	0.03 V pk	0.76 V pk	1.63 V pk
Voltage at 80 % tap	0.02 V pk	0.62 V pk	1.32 V pk
Output power into 50 Ω (ideal steering)	-25 dBm	+1.9 dBm	+8.5 dBm
Carrier feedthrough floor (C_{cb} about 1 pF, Section 10)	about -2 dBm	about -2 dBm	about -2 dBm
Q2 collector minimum (headroom)	5.0 V	4.2 V	3.4 V

The feedthrough floor is a separate, additive term and is not included in the ideal-steering row; it sets the real output at low envelope values and is expected to appear mostly in quadrature, that is as phase error rather than amplitude error at high levels.

Design equations

R_a = R1 + R2 + R3 = 2.33 kΩ (DC path from the PWM to V_C); R_z = R5 ∥ R6 = 415 Ω; R_b = R4 + R_z = 745 Ω; V_{Zth} = 3.3 V · R6 / (R5 + R6) = 0.856 V

$V_C = (R_b \cdot V_{PWM} + R_a \cdot V_{Zth}) / (R_a + R_b) - I_b \cdot (R7 + R_a \parallel R_b) = 0.242 \cdot V_{PWM} + 0.649\ V - I_b \cdot 0.66\ k\Omega$, with V_{PWM} = 3.3 V × duty

$V_C - I_b \cdot (R7 + R_a \parallel R_b) = V_{BE1} + I_{tail} \cdot R8$, with $V_{BE1} = V_T \cdot \ln(I_{tail} / I_S)$ (V_T = 25.9 mV, I_S about 6.7 fA for the 2N3904 model); solved numerically for the table

$I_1 = (2/\pi) \cdot I_{tail}$ (fundamental of a 0 to I_{tail} square wave)

$R_p = R_{in} / p^2$, where R_{in} is the real part of the output-filter input impedance at the tap (about 120 Ω at 200 MHz, not 50 Ω) and p is the tap voltage ratio (about 0.81 for a tap at 80 % of the turns). This gives R_p of about 185 Ω

$Q_L = R_p / (\omega\ L1)$ = 1.5 at 200 MHz, 2.7 at 500 MHz

$f_0 = 1 / (2\pi\sqrt{(L1\ C_{tot})})$: 100 nH with 6.3 pF gives 200.5 MHz; 22 nH with 4.6 pF gives 500.3 MHz. C_{tot} is C5 + TR1 (about 3.4 pF at 200 MHz, about 1.7 pF at 500 MHz), plus device output capacitance and wiring (about 2 pF), plus the net capacitance reflected from the tap and the filter (about 0.9 pF)

Low-pass: 3rd-order π (Butterworth, g = 1, 2, 1, R = 50 Ω). At 260 MHz: C = 12.2 pF, L = 61 nH. At 650 MHz: C = 4.9 pF, L = 24.5 nH

Envelope filter (ladder solved with the R3 to R6 network as its load): DC gain V_C/V_{PWM} = 0.242; −3 dB point about 1.7 kHz; attenuation at 37.5 kHz about 48 dB

3rd harmonic = square-wave 3rd (−9.5 dB) × tank attenuation at $3f_0$ (about −12 dB) × low-pass attenuation at 600 MHz (−22 dB) = about −44 dBc at 200 MHz; the same estimate gives about −47 dBc at 500 MHz

2nd harmonic: zero for an exact 50 % duty at the pair. If the switching threshold (V_{REF} at about 64 % of the GPIO swing) shortens the duty to about 44 % with 1 ns edges, the 2nd-harmonic content of the current is about −15 dB relative to the fundamental; with the tank (−8 dB) and the low-pass (−11.5 dB) this gives about −34 dBc at 200 MHz

Headroom checks pass at all three settings. At full swing the Q2 collector stays about 1.0 V above the common emitter and about 0.1 V above its own base (3.4 V against 3.3 V), and Q1 stays in the active region with V_{CE} of about 0.5 V or more. The tail current drawn from +5 V is at most about 14 mA, so total power is under about 0.1 W.

7. Bill of materials (200 MHz build, all through-hole)

Reference	Value and type	Package	Change for 500 MHz
Q1	2N3904 NPN (tail current source)	TO-92	none

Reference	Value and type	Package	Change for 500 MHz
Q2, Q3	2N5179 NPN, matched pair (RF switch and reference)	TO-72 metal can	BFR91A matched pair (SOT-37); MPSH10 (TO-92) as a cheaper stand-in (see Section 8)
R1, R2	1 kΩ, 1/4 W	axial	none
R3, R4	330 Ω, 1/4 W	axial	none
R5	1.6 kΩ, 1/4 W	axial	none
R6	560 Ω, 1/4 W	axial	none
R7, R12	100 Ω, 1/4 W (R7 base resistor of Q1; R12 dummy load)	axial	none
R8	47 Ω, 1/4 W (tail emitter)	axial	none
R9	100 Ω (RF series)	axial	47 Ω
R10	470 Ω, 1/4 W (reference divider, upper)	axial	none
R11	820 Ω, 1/4 W (reference divider, lower)	axial	none
C1, C2	47 nF, 50 V multilayer	radial	none
C3, C4	100 nF, 50 V multilayer	radial	none
C9	100 nF, 50 V multilayer (5 V decoupling)	radial	add 100 pF NP0 at tank supply node
C10	10 µF, 16 V electrolytic (bulk)	radial	none
C5	1.0 pF NP0 (C0G) tank capacitor	radial	omit
TR1	1–5 pF ceramic trimmer, tank tuning	through-hole trimmer	1–5 pF trimmer (same part)
C6	100 pF NP0 (DC block)	radial	47 pF NP0
C7, C8	12 pF NP0 (low-pass shunts)	radial	4.7 pF NP0
L1	100 nH, tap at 80 % of the winding from the +5 V end (air-core)	through-hole	22 nH, tap at 75 %
L2	62 nH (68 nH acceptable, retune)	through-hole	24 nH
J1	2.54 mm pin header on the Pi GPIO connector; pins 1, 2, 6, 7 and 32 used (drawn in two parts on Sheets 1 and 2)	through-hole	none
J3	SMA female, PCB mount	through-hole	none

Winding guidance. For L1 at 100 nH, start with five turns of 0.8 mm wire on a 6 mm former, stretched to about 8 mm in length, and place the tap at the fourth turn counted from the +5 V end (80 % of the winding). For L2 at 62 nH, start with four turns on the same former, also stretched to about 8 mm. Measure with an LC meter or a VNA and trim the spacing. A commercial axial inductor of the same value is an acceptable substitute for L2; L1 needs a hand-wound tapped coil. All values are starting points.

Pinout note. The schematic uses symbols, not pin numbers. Emitter, base and collector positions differ between the TO-92 (2N3904, MPSH10), TO-72 (2N5179) and SOT-37 (BFR91A) packages, so check each datasheet before soldering. A swapped pair will not steer and will not oscillate in a predictable way; verify with a multimeter first.

8. Changes for 500 MHz on GPIO4

The 200 MHz build can be pushed to 500 MHz with the component changes below. The limiting element is the switch pair. The 2N5179 has a transition frequency f_T of about 0.9 GHz. At 500 MHz that leaves a ratio near 2, which is too low for clean steering edges and makes the output amplitude depend on device speed. Replace Q2 and Q3 with a matched pair of BFR91A transistors, which are leaded SOT-37 devices with a transition frequency of about 5 GHz (check availability, as the part is discontinued by some manufacturers). MPSH10 (TO-92, f_T about 650 MHz) is a lower-cost stand-in. It works, but with less gain and softer edges, and it is marginal at 500 MHz.

Item	200 MHz	500 MHz	Reason
Switch and reference pair Q2, Q3	2N5179 (f_T about 0.9 GHz)	BFR91A matched pair	Device speed must be several times the carrier frequency
RF series resistor R9	100 Ω	47 Ω	Faster base drive and less RC with the device input
Tank inductor L1	100 nH, 80 % tap	22 nH, 75 % tap	Resonance at 500 MHz; the tap sets the full-scale collector swing

Item	200 MHz	500 MHz	Reason
Tank capacitance C5 + TR1	1.0 pF + 1–5 pF	omit C5; TR1 1–5 pF	Resonance including device output capacitance and wiring
DC block C6	100 pF	47 pF	Keeps the reactance small at 500 MHz (about 7 Ω)
Low-pass C7, C8, L2	12 pF, 12 pF, 62 nH (fc 260 MHz)	4.7 pF, 4.7 pF, 24 nH (fc 650 MHz)	Scale the filter to the new carrier
Supply bypass at tank	100 nF	100 nF + 100 pF NP0 at the tank supply node	Low RF impedance on the supply
GPIO4 pad (Pi side)	maximum drive (16 mA)	maximum drive (16 mA), slew limit off (pad control register)	Cleaner carrier edges
Layout and leads	short leads, ground plane recommended	RF leads under 5 mm; solid ground plane under the RF section	Lead inductance is roughly 1 nH per millimetre, a common rule of thumb

At 500 MHz, the expected output is about +8.5 dBm at full scale if the switch devices are fast enough, and the estimated third harmonic is about -47 dBc. Re-tune TR1 for the new resonance and adjust the tap for a full-scale collector swing near 1.7 V peak. The mirrored rpitx documentation lists a frequency range up to 750 MHz; carrier purity and jitter at the top of the range must be measured before relying on it.

9. Bring-up and calibration

1. **DC check without RF.** Confirm V_Z (about 0.74 V with the PWM at 0 %, about 1.16 V at 100 %) and V_{REF} (about 2.10 V with Q3 not conducting, about 2.03 V when it carries the whole tail current). With the PWM at 0 %, V_C should read about 0.65 V and I_{tail} about 0.3 mA. At 100 %, V_C should read about 1.40 V, with about 0.65 V across R8.
2. **Safety first.** Connect a 50 Ω dummy load or a fixed attenuator to the SMA before any RF. Check that your use complies with the licence and spectrum rules that apply where you operate.
3. **Tune the tank.** With the PWM at 100 %, measure the carrier on a spectrum analyser or an SDR. Adjust TR1 for maximum fundamental at the target frequency. Check with a high-impedance probe that the collector of Q2 swings no lower than about 3.3 V.
4. **Measure the transfer curve.** Step the PWM duty from 0 to 100 % in 256 or 512 steps. Log the carrier level at each step with a marker or an SDR power reading, and store the result as a table. Note the level at 0 %: it shows the carrier feedthrough floor.
5. **Invert the curve.** For each desired normalised envelope value A between 0 and 1, look up the PWM duty whose measured output equals A. Apply this lookup in software before each PWM write. The output then follows the envelope in a linear way, down to the feedthrough floor.
6. **Check bandwidth and distortion.** Modulate the envelope with a 100 Hz to 1 kHz tone and inspect the spectrum for sidebands and distortion. Expect the response to roll off above about 1.7 kHz with the filter as drawn.
7. **Check spectral purity.** Measure the second and third harmonics at full scale. If the second is high, check the carrier duty cycle at Q2 and the value of V_{REF}. If the third is above target, confirm the tank Q and the low-pass component values before adding a further filter section.

10. Limitations and safety

Temperature. The base-emitter voltage of Q1 drifts by roughly −2 mV per degree C. At a fixed V_C this changes I_{tail} by about 0.04 mA per degree C, which is about 0.3 % per degree C at full scale and about 0.7 % at mid scale. A larger R8 reduces the drift but also reduces the Q1 headroom, so it needs a higher V_{REF}. The alternative is to replace the resistive reference with one that tracks the drift of Q1.

Efficiency. The complementary current is dissipated in the dummy load R12, so about half of the tail current does not reach the output, and the DC-to-RF efficiency at full scale is about 10 % (about 7 mW out for about 70 mW drawn from +5 V). A balanced output or a transformer would recover that power.

Envelope bandwidth. The default filter gives about 1.7 kHz and attenuates the 37.5 kHz PWM ripple by about 48 dB. That leaves about 2 mV peak at V_C, roughly one 9-bit step, so the effective resolution is about 8 bits. The envelope of a speech SSB signal needs several kilohertz, so the filter as drawn suits only slow envelopes or test tones. Wider EER envelopes need a higher PWM frequency, which means fewer bits, or a different filter. The

usable dynamic range is set by the PWM resolution, the calibration accuracy, the reference drift and the carrier feedthrough described next.

Carrier feedthrough. The collector-base capacitance of Q2 (about 1 pF) couples the GPIO4 swing directly into the tank. At 200 MHz the fundamental of that swing injects about 2.6 mA peak, against 8.8 mA peak of wanted current at full scale, so a carrier of roughly -10 dBc relative to full scale remains when the envelope is zero (about -16 dBc for 0.5 pF). This floor, rather than the PWM resolution, is expected to limit the usable amplitude range to roughly 10 to 15 dB until it is reduced. It must be measured. Lowering the base swing of Q2 (for example with a divider from the base to V_{REF}, using R9 as the series arm) or neutralising the capacitance would reduce it.

Tail-node swing. The common emitter node moves between about 1.2 V (Q3 conducting) and 2.4 V (Q2 conducting) on every cycle. At 200 MHz the capacitance on this node (Q1 collector, junctions and wiring, several picofarads) draws a displacement current of several milliamps, which delays and reshapes the switched current. The $(2/\pi)\cdot I_{tail}$ figure assumes ideal steering and is therefore an upper bound; the output may be lower, and the AM-to-PM conversion higher, than estimated.

Spurious emissions. The harmonic and spur figures are estimates. The second harmonic depends on the duty cycle seen by the pair; for 44 % duty the estimate is about -34 dBc. The output is not compliant with spurious-emission limits without measurement and, possibly, additional filtering.

Operating limits. The total supply current is about 14 mA at +5 V and about 4 mA at +3.3 V. GPIO4 drives only the base of Q2 through 100 Ω (or 47 Ω at 500 MHz), and GPIO12 drives the envelope filter through 1 kΩ, both well within the pad limits.

References

[1] SoapySDR_rpitx repository, github.com/hiperiondev/SoapySDR_rpitx (README, folder structure, repository statistics).

[2] librpitx by F5OEO, github.com/F5OEO/librpitx, as referenced by the SoapySDR_rpitx README.

[3] rpitx README (upstream documentation: carrier range, -c 1 option for GPIO4, RFA amplitude mode), mirrored at github.com/gaionim/rpitx.

[4] Friedt, J.-M. and Courjaud, E., "gr-rpitx: GNU Radio compatible general purpose SDR emitter using the Raspberry Pi(4) internal phase locked loop", GNU Radio Conference 2021, pubs.gnuradio.org/index.php/grcon/article/view/107.

[5] Kahn, L. R., "Single Sideband Transmission by Envelope Elimination and Restoration", Proc. IRE, vol. 40, no. 7, pp. 803–806, 1952.

[6] Broadcom BCM2835 ARM Peripherals datasheet, pad control registers. The drive-field mapping should be verified for your SoC revision.